



## **API Reference**

# SDN Agent to KMS API - KMS Server

**API Version: 2.2.0**

Messages the SDN sends to the KMS

### **CONTACT**

**URL:** <https://github.com/ait-crypto/kms-sdn-agent-interface-specification/discussions>

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# Security and Authentication

## SECURITY SCHEMES

| KEY   | TYPE      | DESCRIPTION                                                                    |
|-------|-----------|--------------------------------------------------------------------------------|
| https | mutualTLS | Connections must use TLS 1.3. The server enforces this at the transport layer. |

# API

## 1. API

Generic endpoints related with this API

### 1.1 GET /version

#### Get this API version

QUICKS API version number which is implemented and supported by the KMS

### REQUEST

No request parameters

### RESPONSE

**STATUS CODE - 200:** Provided API version

#### RESPONSE MODEL - application/json

| NAME                        | TYPE   | DESCRIPTION                                                                                                                                                                                                                                      |
|-----------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                                                                                                                                                                                                                                  |
| version*                    | string | <b>PATTERN:</b> ^(0 [1-9]\d*)\.(0 [1-9]\d*)\.(0 [1-9]\d*)(?:-((?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)*)?)?(?:\+(0 [1-9]\d*[a-zA-Z-][0-9a-zA-Z-]*)?)?)?\$<br>Version number in semantic versioning |

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 2. CONTROL

Generic control endpoints

### 2.1 PUT /control/status

#### Modify state of the KMS

Modify state of the KMS, it can be online, stopped or offline.

#### REQUEST

##### REQUEST BODY - application/json

| NAME   | TYPE | DESCRIPTION                                                          |
|--------|------|----------------------------------------------------------------------|
| state* | enum | <b>ALLOWED:</b> stopped, offline, online<br>Target state of the KMS. |

#### RESPONSE

**STATUS CODE - 200:** changing state successful

**STATUS CODE - 400:** Bad request

##### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 403:** Forbidden, client does not have the required rights.

##### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

##### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE   | DESCRIPTION                              |
|------------------------------------|--------|------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |        |                                          |
| type                               | string | Webpage with information on this problem |
| title*                             | string | Brief description of the problem         |

| NAME    | TYPE    | DESCRIPTION                                             |
|---------|---------|---------------------------------------------------------|
| status* | integer | Response code or status code                            |
| detail* | string  | Information on the problem and hints on how to solve it |

## 2.2 PUT /control/qkd-device/{device\_id}

### Update an existing QKD device

Update mutable fields of a known QKD device.

## REQUEST

### PATH PARAMETERS

| NAME       | TYPE | EXAMPLE | DESCRIPTION |
|------------|------|---------|-------------|
| *device_id | uuid |         |             |

### REQUEST BODY - application/json

| NAME              | TYPE    | DESCRIPTION                                                                                    |
|-------------------|---------|------------------------------------------------------------------------------------------------|
| link_id           | string  | The ID of the link this device is associate to.                                                |
| label             | string  | Human readable label, for example vendor name and model number.                                |
| state             | enum    | <b>ALLOWED:</b> online, stopped, offline<br>Current or target state after successful operation |
| interface         | enum    | <b>ALLOWED:</b> etsi_004, etsi_014, skip<br>Interface type of the QKD device                   |
| protocol          | string  | Transport protocol used, for ETSI GS QKD 014 typically https.                                  |
| peer_kms          | string  | to which peer is this device connected, does not have to be UUID format, but unique in network |
| server_address    | string  | URI of the QKD device server. Specify domain (or IP) and port.                                 |
| <b>etsi_014</b>   | object  |                                                                                                |
| primary_id*       | string  | In ETSI GS QKD 014 also called Master SAE ID                                                   |
| secondary_id*     | string  | In ETSI GS QKD 014 also called Slave SAE ID                                                    |
| <b>etsi_004</b>   | object  |                                                                                                |
| source*           | string  | source URI                                                                                     |
| destination*      | string  | source URI                                                                                     |
| key_chunk_size    | integer | length of key chunks in Byte.                                                                  |
| max_bps           | integer |                                                                                                |
| min_bps           | integer |                                                                                                |
| jitter            | integer |                                                                                                |
| priority          | integer |                                                                                                |
| timeout           | integer |                                                                                                |
| ttl               | integer |                                                                                                |
| metadata_mimetype | string  |                                                                                                |

## RESPONSE

**STATUS CODE - 200:** QKD device updated successfully

**STATUS CODE - 400:** Invalid input (missing device\_id or malformed update object)

**RESPONSE MODEL - application/json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found. This corresponds to the endpoint not found, as 404s can also be generated by the webserver. Use 400 if the device ID string was unknown to KMS

**STATUS CODE - 500:** Server-side error while updating the device

**RESPONSE MODEL - application/json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 2.3 POST /control/qkd-device/{device\_id}

### Register a new QKD device

Create/register a new QKD device in the KMS instance.

## REQUEST

**PATH PARAMETERS**

| NAME       | TYPE | EXAMPLE | DESCRIPTION |
|------------|------|---------|-------------|
| *device_id | uuid |         |             |

**REQUEST BODY - application/json**

| NAME            | TYPE   | DESCRIPTION                                                                                    |
|-----------------|--------|------------------------------------------------------------------------------------------------|
| link_id*        | string | The ID of the link this device is associate to.                                                |
| label*          | string | Human readable label, for example vendor name and model number.                                |
| state*          | enum   | <b>ALLOWED:</b> online, stopped, offline<br>Current or target state after successful operation |
| interface*      | enum   | <b>ALLOWED:</b> etsi_004, etsi_014, skip<br>Interface type of the QKD device                   |
| protocol*       | string | Transport protocol used, for ETSI GS QKD 014 typically https.                                  |
| peer_kms*       | string | to which peer is this device connected, does not have to be UUID format, but unique in network |
| server_address* | string | URI of the QKD device server. Specify domain (or IP) and port.                                 |
| <b>etsi_014</b> | object |                                                                                                |
| primary_id*     | string | In ETSI GS QKD 014 also called Master SAE ID                                                   |

| NAME              | TYPE    | DESCRIPTION                                 |
|-------------------|---------|---------------------------------------------|
| secondary_id*     | string  | In ETSI GS QKD 014 also called Slave SAE ID |
| <b>etsi_004</b>   | object  |                                             |
| source*           | string  | source URI                                  |
| destination*      | string  | source URI                                  |
| key_chunk_size    | integer | length of key chunks in Byte.               |
| max_bps           | integer |                                             |
| min_bps           | integer |                                             |
| jitter            | integer |                                             |
| priority          | integer |                                             |
| timeout           | integer |                                             |
| ttl               | integer |                                             |
| metadata_mimetype | string  |                                             |

## RESPONSE

**STATUS CODE - 201:** QKD device successfully created

**STATUS CODE - 400:** Invalid input (missing fields or malformed body)

### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 409:** Device already exists (conflict on device\_id)

### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Server-side error while creating the device

### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 2.4 DELETE /control/qkd-device/{device\_id}

### Delete QKD device

Remove connected QKD device, KMS will no longer try to obtain keys from it.

## REQUEST

### PATH PARAMETERS

| NAME       | TYPE | EXAMPLE | DESCRIPTION |
|------------|------|---------|-------------|
| *device_id | uuid |         |             |

## RESPONSE

**STATUS CODE - 204:** QKD device correctly deleted

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist.

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 3. LINK

Monitoring and control related with the link

### 3.1 GET /link/performance/{link\_id}

#### Get performance data for link

Get the performance data for the link with the given link id. Query parameter for entries is optional.

#### REQUEST

##### PATH PARAMETERS

| NAME     | TYPE | EXAMPLE | DESCRIPTION |
|----------|------|---------|-------------|
| *link_id | uuid |         | link's UUID |

##### QUERY PARAMETERS

| NAME    | TYPE    | EXAMPLE | DESCRIPTION                                                                                                                                     |
|---------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| entries | integer |         | Optional. Number of history records requested. If not specified maximal available history is returned. If 0, only current value is transmitted. |

#### RESPONSE

**STATUS CODE - 200:** The response contains KMS performance data

##### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                                                                                                        |
|------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                                                                                                    |
| <b>current*</b>                    | object  |                                                                                                                                    |
| timestamp*                         | string  | ISO 8601 format                                                                                                                    |
| delta_t*                           | integer | <b>DEFAULT:10</b><br>Time period over which keys are counted for the rate calculation. The unit is milliseconds.                   |
| IKR*                               | integer | Internal Key Rate = amount of key material reserved for internal use during a delta_t period divided by delta_t. Given in Byte/sec |
| EXKR*                              | integer | External Key Rate = amount of key material reserved for external use during a delta_t period divided by delta_t. Given in Byte/sec |
| SKR*                               | integer | Secret Key Rate = amount of key material received from QKD devices during a delta_t period divided by delta_t. Given in Byte/sec   |
| KAV*                               | integer | Key Availability = amount of key material in Byte available to Applications during a delta_t period                                |
| <b>history</b>                     | array   |                                                                                                                                    |
| timestamp*                         | string  | ISO 8601 format                                                                                                                    |
| delta_t*                           | integer | <b>DEFAULT:10</b><br>Time period over which keys are counted for the rate calculation. The unit is milliseconds.                   |

| NAME  | TYPE    | DESCRIPTION                                                                                                                        |
|-------|---------|------------------------------------------------------------------------------------------------------------------------------------|
| IKR*  | integer | Internal Key Rate = amount of key material reserved for internal use during a delta_t period divided by delta_t. Given in Byte/sec |
| EXKR* | integer | External Key Rate = amount of key material reserved for external use during a delta_t period divided by delta_t. Given in Byte/sec |
| SKR*  | integer | Secret Key Rate = amount of key material received from QKD devices during a delta_t period divided by delta_t. Given in Byte/sec   |
| KAV*  | integer | Key Availability = amount of key material in Byte available to Applications during a delta_t period                                |

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist, probably invalid link\_id given.

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 3.2 PUT /link/relay/{key\_stream\_id}

### Update a link relay configuration

Replace a link relay configuration with the data in the request, using the given key\_stream\_id as an identifier. Only the data to be changed should be included in data body, supporting partial updates.

## REQUEST

#### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION          |
|----------------|------|---------|----------------------|
| *key_stream_id | uuid |         | key_stream_id's UUID |

#### REQUEST BODY - application/json

| NAME        | TYPE   | DESCRIPTION            |
|-------------|--------|------------------------|
| source      | string | Source APP's URI or ID |
| destination | array  |                        |

| NAME                  | TYPE   | DESCRIPTION                                                                                                                                                                                                                             |
|-----------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| next_node_info        | array  |                                                                                                                                                                                                                                         |
| qos_request           | object |                                                                                                                                                                                                                                         |
| peer_application_info | array  |                                                                                                                                                                                                                                         |
| kms_id                | string | Unique ID of the KMS instance. If the KMS_ID is the same as the KMS, which sent the request, the applications are served by the caller KMS. A potential use-case for that are group keys, where peer applications are on the same node. |
| application_ids       | array  |                                                                                                                                                                                                                                         |

## RESPONSE

**STATUS CODE - 200:** The relay configuration associated with the given key\_stream\_id has been replaced by the data in the request

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist, probably invalid key\_stream\_id given.

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 3.3 POST /link/relay/{key\_stream\_id}

### Create link relay configuration

Create a link relay configuration record with the data in the request, using the given key\_stream\_id as an identifier

## REQUEST

### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION          |
|----------------|------|---------|----------------------|
| *key_stream_id | uuid |         | key_stream_id's UUID |

### REQUEST BODY - application/json

| NAME                  | TYPE        | DESCRIPTION                                                                                                                                                                                                                             |
|-----------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| source*               | string      | URI or ID of the source APP                                                                                                                                                                                                             |
| destination*          | array       |                                                                                                                                                                                                                                         |
| next_node_info*       | array       |                                                                                                                                                                                                                                         |
| qos_request           | {recursive} | qos-data                                                                                                                                                                                                                                |
| peer_application_info | array       |                                                                                                                                                                                                                                         |
| kms_id                | string      | Unique ID of the KMS instance. If the KMS_ID is the same as the KMS, which sent the request, the applications are served by the caller KMS. A potential use-case for that are group keys, where peer applications are on the same node. |
| application_ids       | array       |                                                                                                                                                                                                                                         |

## RESPONSE

**STATUS CODE - 201:** A new relay configuration has been created with the given key\_stream\_id as an identifier, from the data in the request

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 3.4 DELETE /link/relay/{key\_stream\_id}

### Delete relay configuration

Delete the relay configuration associated with the given key\_stream\_id

## REQUEST

### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION       |
|----------------|------|---------|-------------------|
| *key_stream_id | uuid |         | key stream's UUID |

## RESPONSE

**STATUS CODE - 204:** Forward table row deleted

**STATUS CODE - 400:** Bad request

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist, probably invalid key\_stream\_id given.

**STATUS CODE - 500:** Internal error

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4. MONITOR

Generic monitoring endpoints

### 4.1 GET /monitor/capabilities

#### Get KMS capabilities

Query which essential features are supported by this instance.

#### REQUEST

No request parameters

#### RESPONSE

**STATUS CODE - 200:** The response states different KMS API capabilities.

##### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                                                                                                                                                                                                                     |
|------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                                                                                                                                                                                                                 |
| key_relay_support*                 | enum    | <b>ALLOWED:</b> true, false<br>Is this KMS instance capable of relaying keying material? If true, key relay supported, if false it isn't.                                                                                                       |
| app_api_support*                   | array   |                                                                                                                                                                                                                                                 |
| etsi_014_info                      | object  |                                                                                                                                                                                                                                                 |
| kms_id*                            | string  | This KMS instance ID                                                                                                                                                                                                                            |
| key_size*                          | integer | $\geq 0$<br>Number in bit stating the default key size of this KMS, all other parameters use this for calculations.                                                                                                                             |
| stored_key_count*                  | integer | $\geq 0$<br>Number of stored keys which are available and could be delivered to the application. Number of keys is based on number of entries of the default key size. Sum of all Links, per link data available via link performance endpoint. |
| max_key_count*                     | integer | $\geq 0$<br>Number stating this KMS instance maximum key store. This assumes above default key size.                                                                                                                                            |
| max_key_per_request*               | integer | $\geq 0$<br>Number stating this KMS instance maximum key entries, which can be handled at once. For example number of keys which can be requested via ETSI GS QKD 014 at once or forwarded through this instance at once.                       |
| max_key_size*                      | integer | $\geq 0$<br>Number in bit stating the biggest key size this KMS can handle.                                                                                                                                                                     |
| min_key_size*                      | integer | $\geq 0$<br>Number in bit stating the smallest key size this KMS can handle.                                                                                                                                                                    |
| max_sae_id_count*                  | integer | $\geq 0$<br>Number stating if this KMS supports group keys and if so with how many participants. 0 means no group keys, 1 is one additional slave SAE etc.                                                                                      |
| status_extension                   | object  |                                                                                                                                                                                                                                                 |

**STATUS CODE - 400:** Bad request

## RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

## RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4.2 GET /monitor/device-info

### Get vendor, model and version

Publish KMS instance vendor, model and version

## REQUEST

No request parameters

## RESPONSE

**STATUS CODE - 200:** The response contains the vendor, device and version

## RESPONSE MODEL - application/json

| NAME                        | TYPE   | DESCRIPTION                                                                                                                                                                                                                                      |
|-----------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                                                                                                                                                                                                                                  |
| vendor*                     | string | Vendor of the KMS                                                                                                                                                                                                                                |
| model*                      | string | Device brand name (model)                                                                                                                                                                                                                        |
| version*                    | string | <b>PATTERN:</b> ^(0 [1-9]\d*)\.(0 [1-9]\d*)\.(0 [1-9]\d*)(?:-((?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?))?(?:\+([0-9a-zA-Z-]+(?:\.[0-9a-zA-Z-]+)*)?)?)?\$<br>Version number in semantic versioning |

**STATUS CODE - 400:** Bad request

## RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4.3 GET /monitor/status

### Get KMS status

Get the status from the KMS, it can be online or stopped.

## REQUEST

### QUERY PARAMETERS

| NAME                     | TYPE    | EXAMPLE | DESCRIPTION                                                                                                 |
|--------------------------|---------|---------|-------------------------------------------------------------------------------------------------------------|
| new_operation_state_only | boolean |         | If true, only consider operational status events since last query. If not given default of "false" is used. |

## RESPONSE

**STATUS CODE - 200:** description of the status

**RESPONSE MODEL - application/json**

| NAME                        | TYPE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |      |                                                                                                                                                                                                                                                                                                                                                          |
| state*                      | enum | <b>ALLOWED:</b> online, stopped, offline<br>State of the KMS. "online" if operational, stopped if running but not operational, offline if not operational. If it can't answer it is assumed to be offline, see error below. Stopped is a state, where the KMS is running but not operational, which could for example be a maintenance state or similar. |
| operation                   | enum | <b>ALLOWED:</b> normal operation, warnings, errors, critical<br>state if there are any operational issues. This may correlate to entries in the KMS log.                                                                                                                                                                                                 |

**STATUS CODE - 400:** Bad request

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** KMS can't be reached, therefore has to be assumed offline.

**STATUS CODE - 500:** Internal error

## RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4.4 GET /monitor/log

### Get full logging info

get the log file from the KMS

## REQUEST

No request parameters

## RESPONSE

**STATUS CODE - 200:** Log file of the KMS. State the file format in the MIME type notation in the http Content-Type header. Any type can be returned, some examples are given, but it is beyond this specification to specify the log format.

### RESPONSE MODEL - \*/\*

#### EXAMPLE:

```
"[File content of any type]"
```

### RESPONSE MODEL - text/plain

### RESPONSE MODEL - application/json

### RESPONSE MODEL - application/vnd.sqlite3

### RESPONSE MODEL - application/octet-stream

**STATUS CODE - 400:** Bad request

## RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

## RESPONSE MODEL - application/problem+json

| NAME                        | TYPE   | DESCRIPTION                              |
|-----------------------------|--------|------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                          |
| type                        | string | Webpage with information on this problem |
| title*                      | string | Brief description of the problem         |

| NAME    | TYPE    | DESCRIPTION                                             |
|---------|---------|---------------------------------------------------------|
| status* | integer | Response code or status code                            |
| detail* | string  | Information on the problem and hints on how to solve it |

## 4.5 GET /monitor/qkd-device

### Get known QKD devices and status

Receive QKD device information the KMS instance has.

#### REQUEST

No request parameters

#### RESPONSE

**STATUS CODE - 200:** QKD devices information correctly returned

##### RESPONSE MODEL - application/json

| NAME                               | TYPE   | DESCRIPTION                                                                                    |
|------------------------------------|--------|------------------------------------------------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |        |                                                                                                |
| <b>qkd_devices</b>                 | array  |                                                                                                |
| device_id*                         | string |                                                                                                |
| link_id*                           | string | The ID of the link this device is associate to.                                                |
| label                              | string | Human readable label, for example vendor name and model number.                                |
| state*                             | enum   | <b>ALLOWED:</b> online, stopped, offline<br>Current or target state after successful operation |
| interface*                         | enum   | <b>ALLOWED:</b> etsi_004, etsi_014, skip<br>Interface type of the QKD device                   |
| protocol*                          | string | Transport protocol used, for ETSI GS QKD 014 typically https.                                  |
| peer_kms*                          | string | to which peer is this device connected, does not have to be UUID format, but unique in network |

**STATUS CODE - 400:** Bad request

##### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** QKD devices information could not be returned because of a server-side error

##### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                              |
|------------------------------------|---------|------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                          |
| type                               | string  | Webpage with information on this problem |
| title*                             | string  | Brief description of the problem         |
| status*                            | integer | Response code or status code             |

| NAME    | TYPE   | DESCRIPTION                                             |
|---------|--------|---------------------------------------------------------|
| detail* | string | Information on the problem and hints on how to solve it |